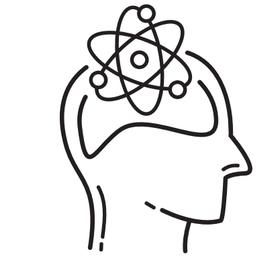


Generating Design Solutions

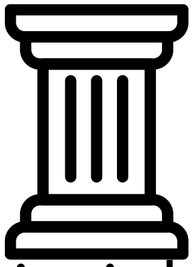
Guidelines, Theories and Principles

Guidelines, Principles, and Theories

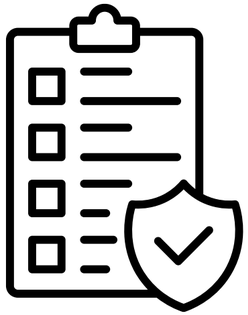
- **Guidelines**: Low-level focused advice about good practices and cautions against dangers
- **Principles**: Mid-level strategies or rules to analyze and compare design alternatives
- **Theories**: High-level widely applicable frameworks to draw on during design and evaluation, as well as to support communication and teaching



Theories



Principles



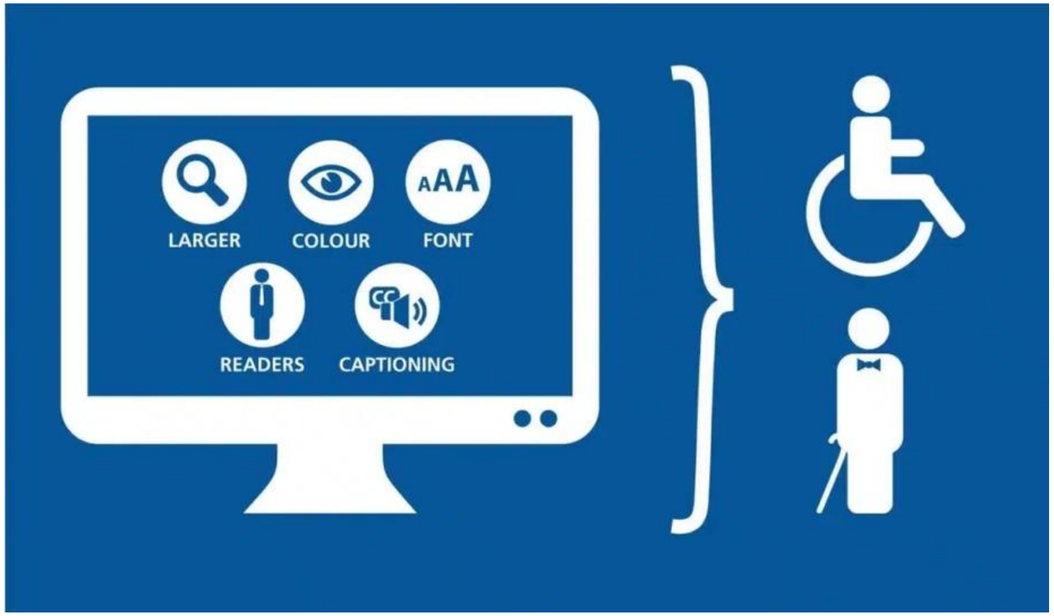
Guidelines

guidelines

- Guidelines provide structured advice and rules that designers should follow to ensure their solutions meet specific **standards** or norms. These are often based on extensive research, **best practices**, or legal requirements, and they help in creating systems that are consistent, usable, and safe.



- Web Accessibility Guidelines (WCAG)
  - Ensure that websites are usable by people with disabilities.

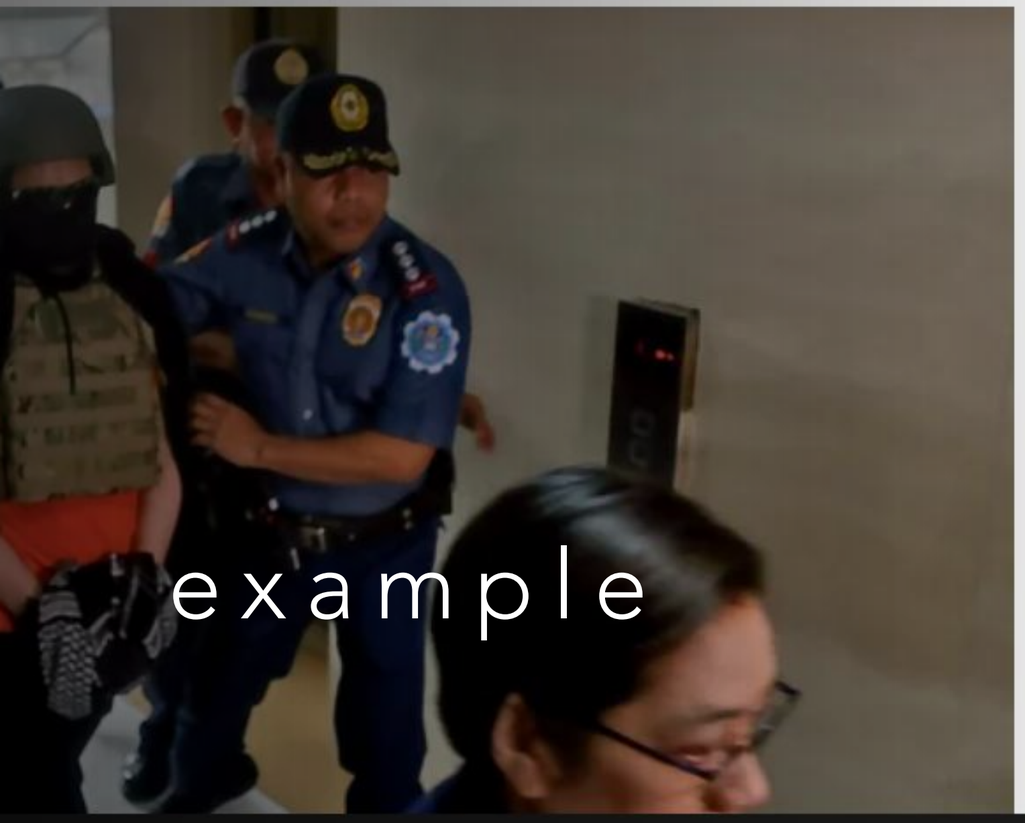


- Provide text alternatives for non-text content like images or videos (e.g., adding ALT text to images).
- Make content accessible through keyboard navigation, ensuring users who cannot use a mouse can still interact with the site.
- Design for readability, ensuring there is sufficient color contrast between text and background, and that the text is legible.

n.com

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World

Harris, Trump campaign in battlegrounds as migrant row intensifies

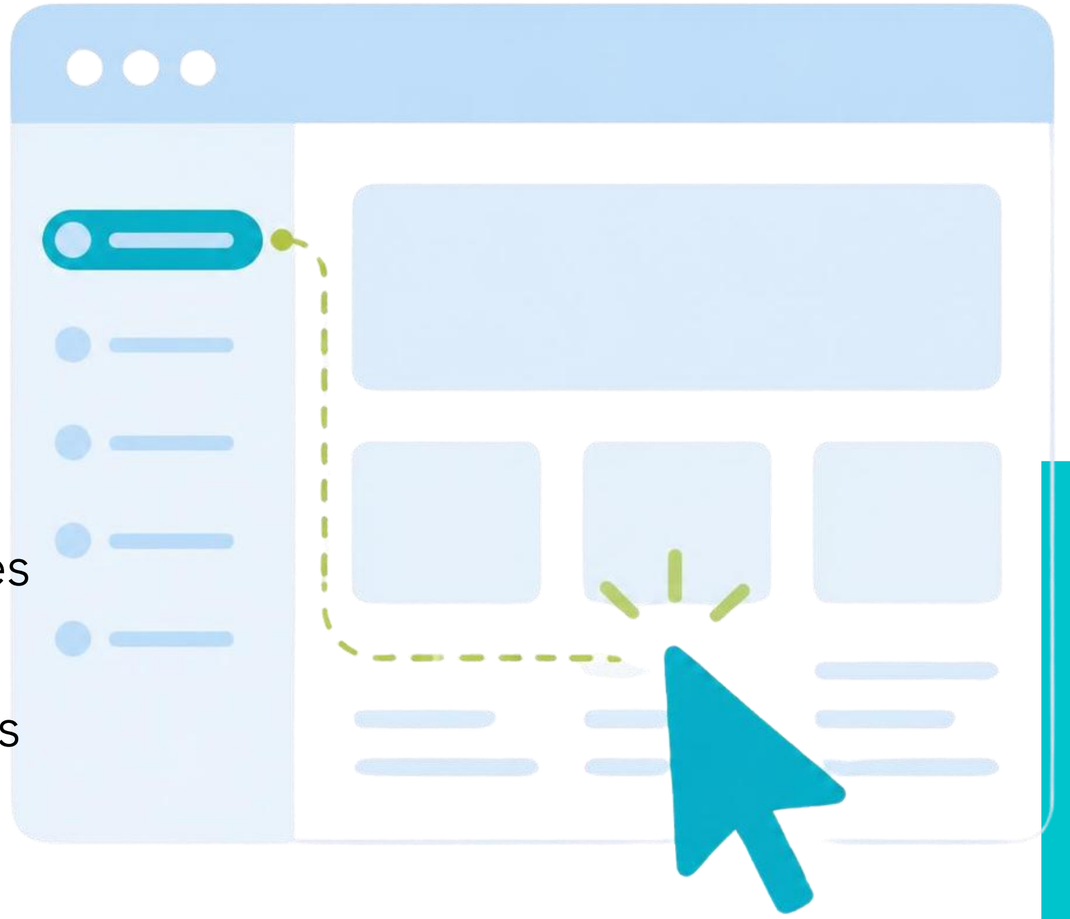
2 hours ago

12 hours

- For a news website, guidelines would mandate that all visual elements (images, graphs) have text descriptions so that visually impaired users can understand the content through screen readers.

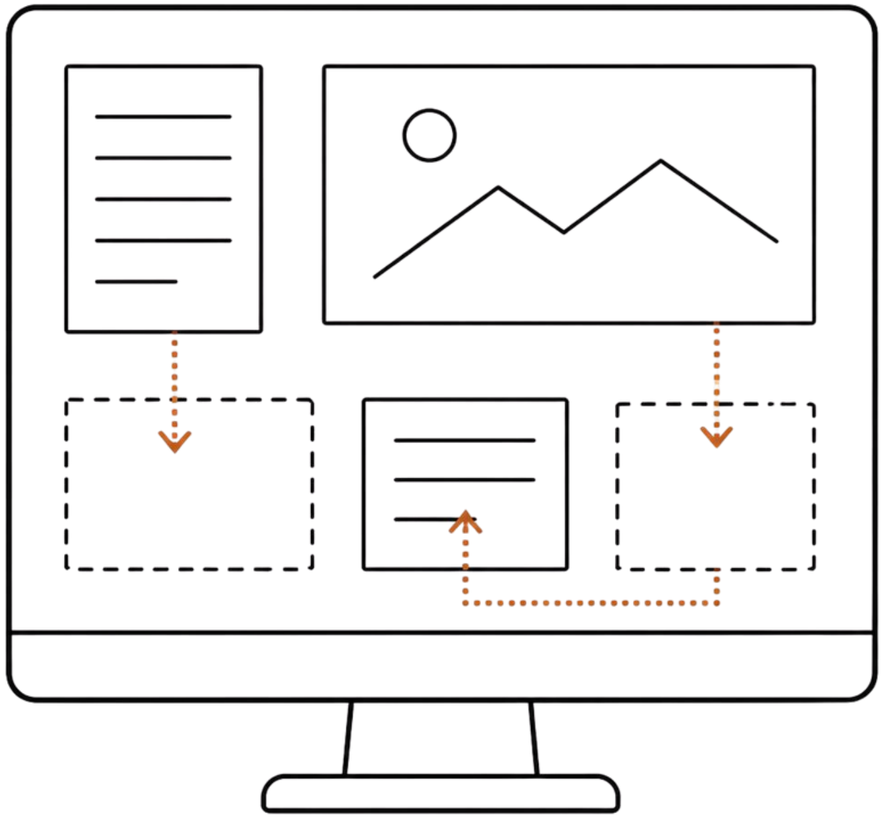
Navigating the interface

- Standardize task sequences
- Ensure that embedded links are descriptive
- Use unique and descriptive headings
- Use check boxes for binary choices
- Use radio buttons for mutually exclusive choices
- Develop pages that will print properly
- Use thumbnail images to preview larger images



Organizing the display

- Smith and Mosier (1986) offer five high-level goals
  - Consistency of data display
  - Efficient information assimilation by the user / minimal input actions by user
  - Minimal memory load on the user
  - Compatibility of data display with data entry
  - Flexibility for user control of data display



Getting the user's attention

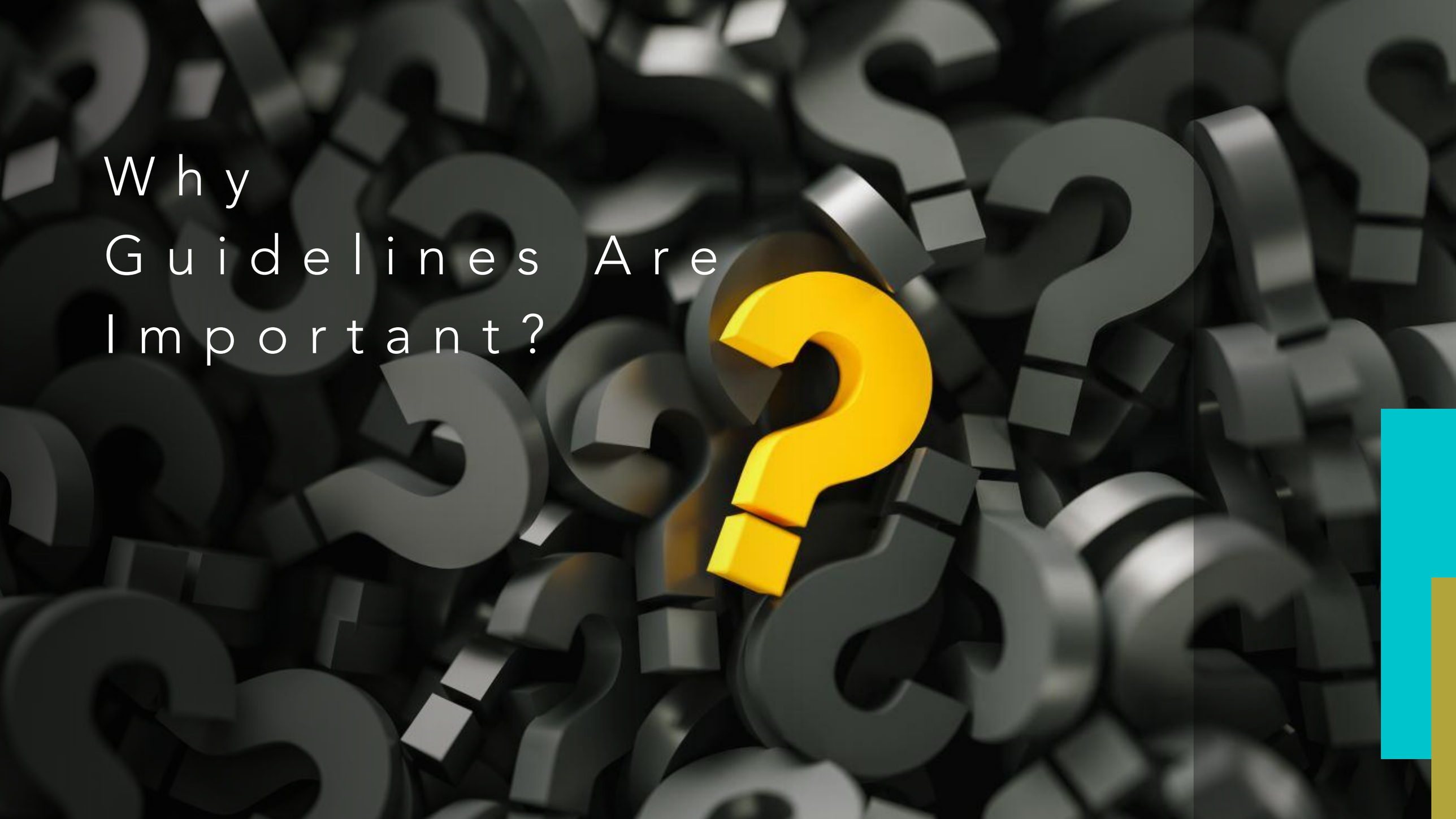


- **Visual Hierarchy & Contrast**
  - Intensity - use two levels only, with limited use of high intensity
  - Size – use up to 4 sizes
  - Choice of fonts – use up to 3 fonts
- **Attention-Grabbing & Marking Techniques**
  - Marking – various ways: underline the item, enclose it in a box, point it with an arrow, use an asterisk, bullet, dash, + or X sign
  - Blinking – use this method cautiously [2-4 Hz]
- **Color, Audio, & Motion**
  - Color – use up to 4 standard colors
  - Animation – use it only when meaningful
  - Audio – soft tones [regular] vs. harsh tones [emergency]
- Avoid cluttered screens, don't overuse the above

Facilitate data entry

- **Consistency of data-entry transactions** :Similar sequences of actions speed up learning
- **Minimal input actions by user:** Fewer input actions mean greater operator productivity, and usually less error; avoid redundant data entry
- **Minimal memory load on users:** Users should not be required to remember lengthy lists of codes
- **Compatibility of data entry with data display:** The format of data-entry information should be linked closely to the format of displayed information, such as dashes in telephone numbers
- **Flexibility for user control of data entry:** Experienced users prefer to enter information in a sequence that they can control

Why Guidelines Are Important?





Activity #1: Guidelines Audit

- Instruction
  - Choose a webpage or app to review.
  - Review it against Web Accessibility (WCAG) and other guidelines presented. Check for missing text alternatives, poor color contrast, unclear links, inconsistent navigation, and inadequate focus indicators.
  - List every issue you find and categorize it (Accessibility / Navigation / Display, etc.).
  - Propose a specific fix for each issue.
  - Share your findings with the class and discuss which guidelines were most frequently violated.

ACTIVITY #2:Inside the Guidelines: Decoding Modern UI Systems

- Instructions
  - Select 1 from among these websites which implements design guidelines
    - https://webstyleguide.com/
    - https://developer.apple.com/design/human-interface-guidelines/
    - https://www.w3.org/WAI/standards-guidelines/wcag/
    - https://digital.gov/resources/mobile-user-experience-guidelines/
    - https://design-system.service.gov.uk/
    - https://www.microsoft.com/design/fluent/
    - https://material.io/
  - Create a comprehensive outline of specific guidelines included on the website
  - Make a presentation

.

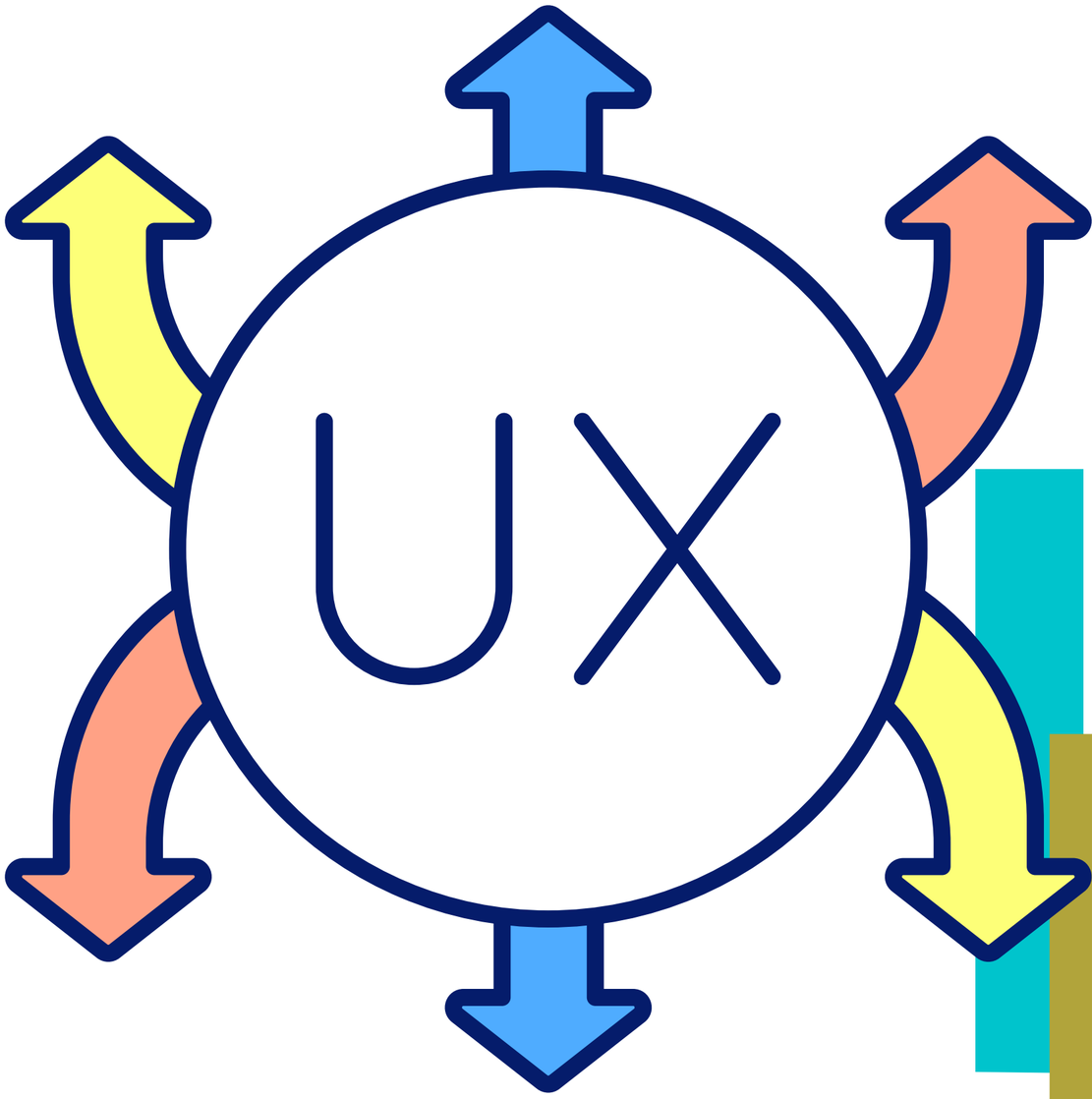
Principles

- Principles are fundamental truths or laws that guide the design process
- It ensures that the solution is not only usable but also effective, aesthetically pleasing, and functional.
- Principles are broader than guidelines and often apply universally across different contexts and mediums.



Principles of User Experience (UX) Design

- Clarity
  - Ensure that the design is easy to understand and navigate
- User Control
  - Allow users to feel in control by providing undo options, easily accessible help, and ability to customized elements



Visual-design principles inform us how design elements go together to create well-rounded and thoughtful visuals. Graphics that take advantage of the principles of good visual design can drive engagement and increase usability.

SCALE

The principle of scale refers to using relative size to signal importance and rank in a composition.



VISUAL HIERARCHY

The principle of visual hierarchy refers to guiding the eye on the page so that it attends to design elements in the order of their importance.



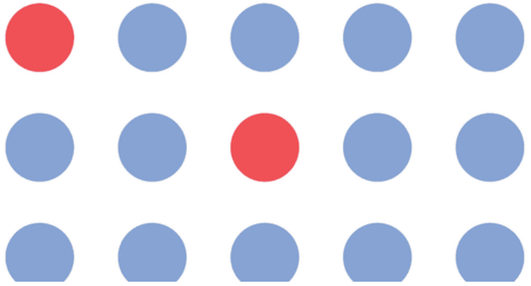
BALANCE

Balance occurs when there is an equally distributed amount of visual signal on both sides of an imaginary axis.



CONTRAST

The principle of contrast refers to the juxtaposition of visually dissimilar elements in order to convey the fact that these elements are different.



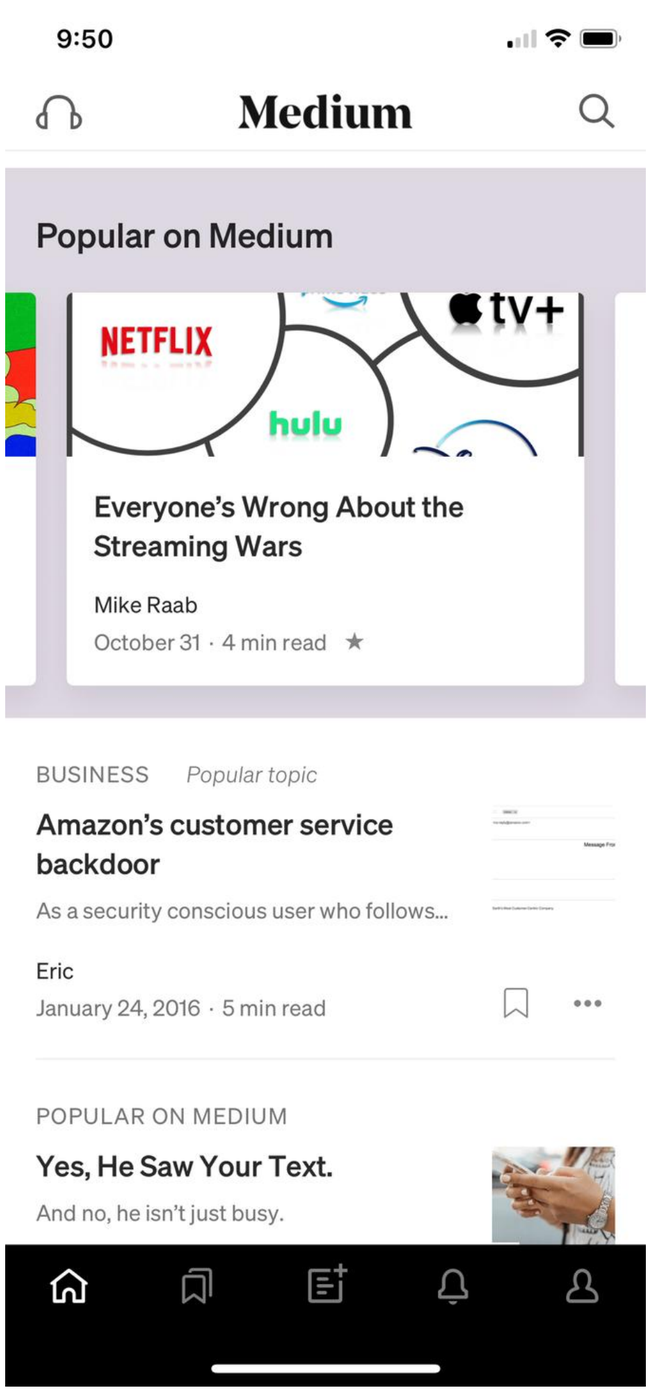
GESTALT PRINCIPLES

Gestalt principles capture our tendency to perceive the whole as opposed to the individual elements.

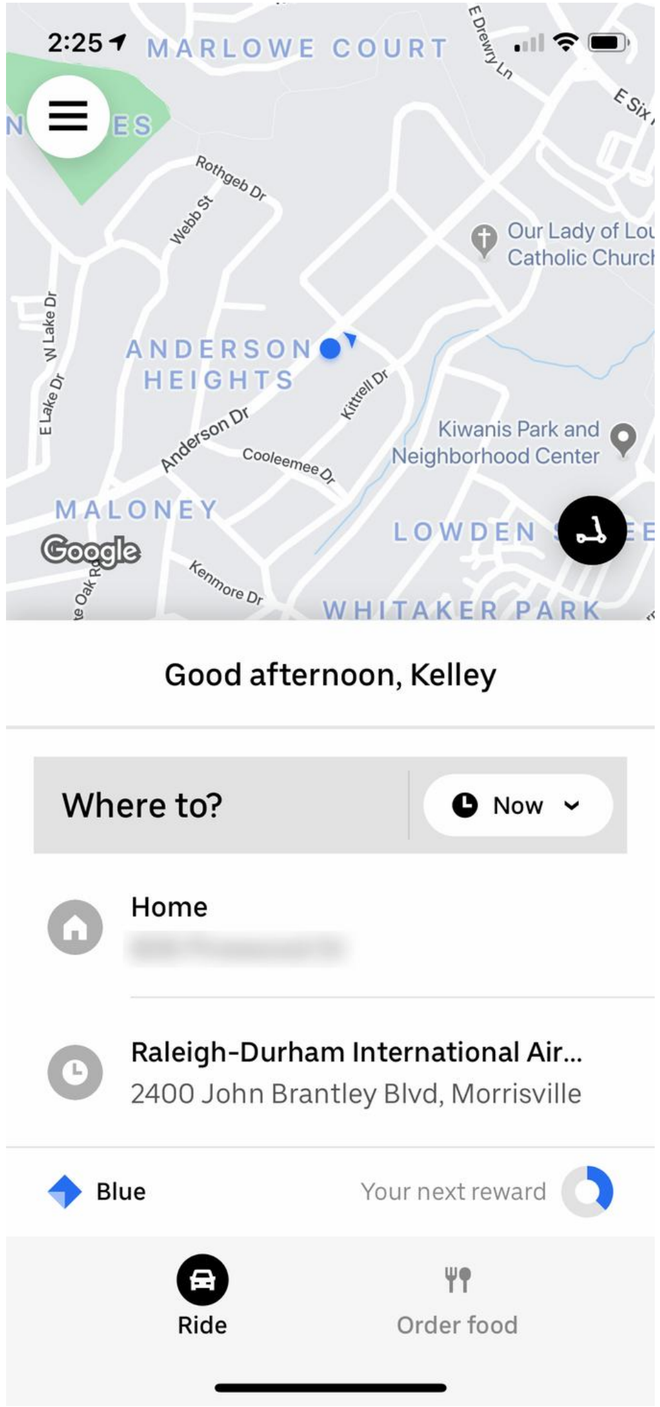


source: https://www.nngroup.com/articles/principles-visual-design/

The principle of scale



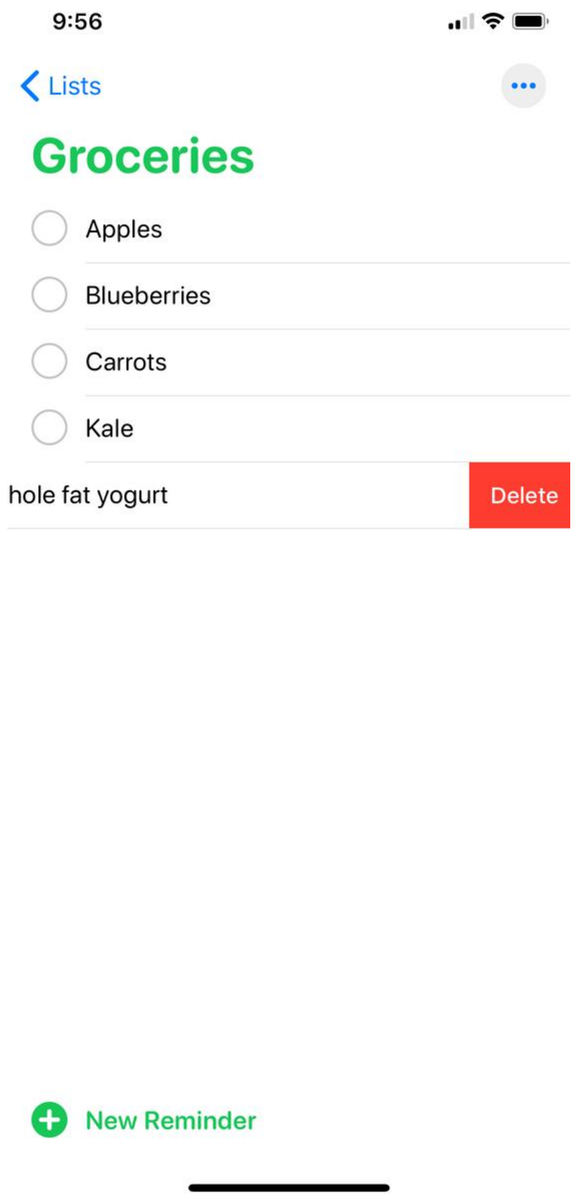
The principle of visual hierarchy



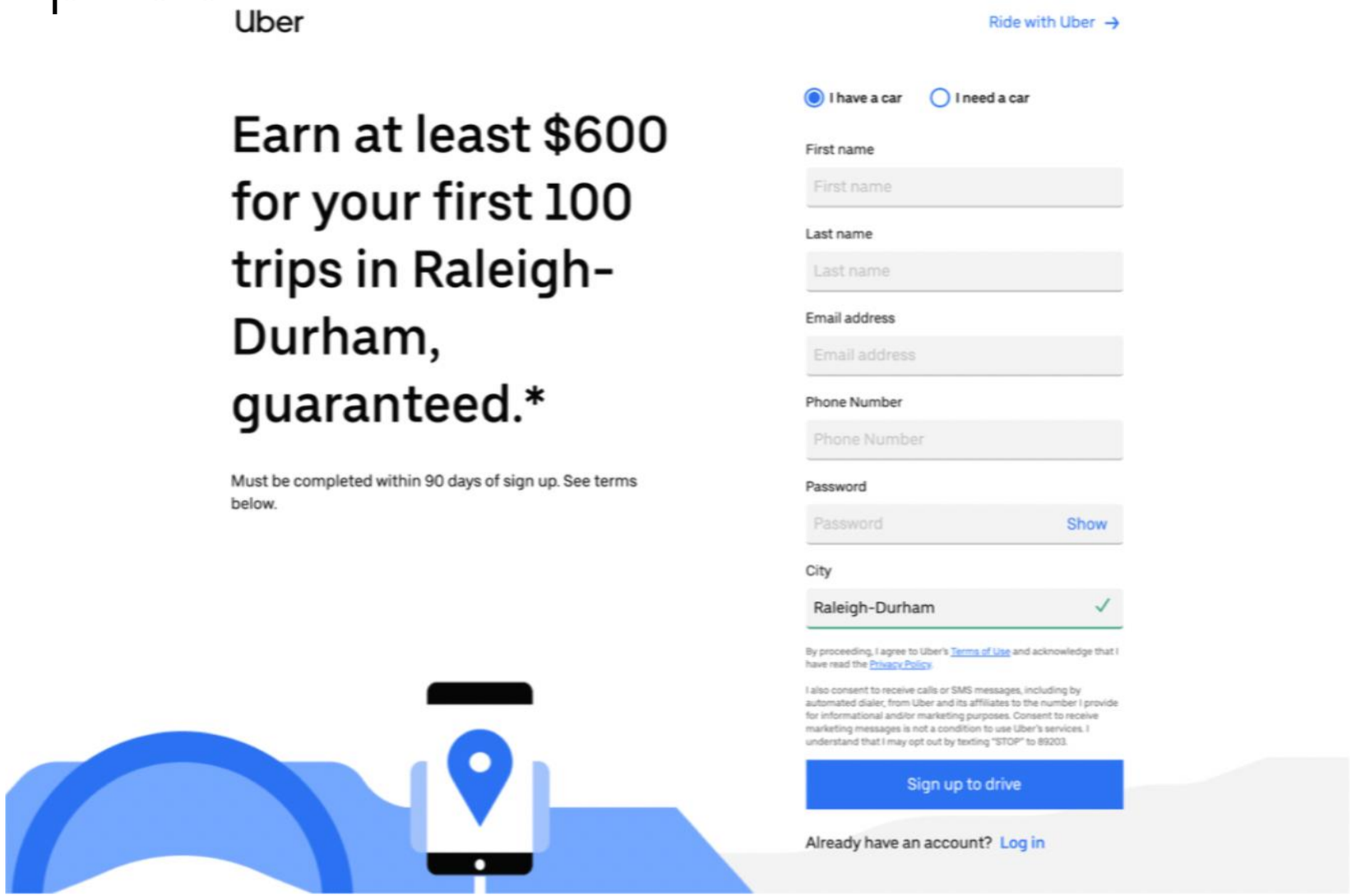
The principle of balance



The principle of contrast



Gestalt principles



Fundamental principles

- Determine user's skill levels
- Identify the tasks
- Choose between five primary interaction styles
- Apply 8 golden rules of interface design
- Prevent errors
- Ensure human control while increasing automation

Determine the user's skill levels

- "Know thy user"
- Age, gender, physical and cognitive abilities, education, cultural or ethnic background, training, motivation, goals and personality
- Design goals based on skill level
  - Novice or first-time users
  - Knowledgeable intermittent users
  - Expert frequent users
- Multi-layer designs, personalized menus, control of informative feedback

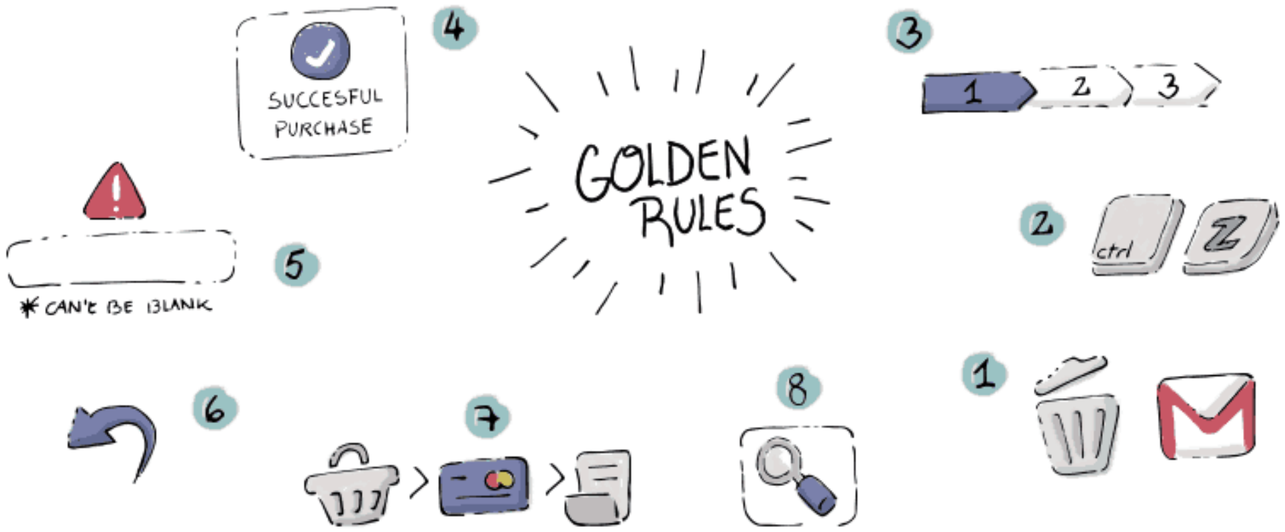
Choose an interaction style

- Direct manipulation
- Navigation and Menu selection
- Form fill-in
- Command language
- Natural language

Advantages	Disadvantages
Direct manipulation	
Visually presents task concepts	May be hard to program
Allows easy learning	May require graphics display and pointing devices
Allows easy retention	
Allows errors to be avoided	
Encourages exploration	
Affords high subjective satisfaction	
Menu selection	
Shortens learning	Presents danger of many menus
Reduces keystrokes	May slow frequent users
Structures decision making	Consumes screen space
Permits use of dialog-management tools	Requires rapid display rate
Allows easy support of error handling	
Form fill-in	
Simplifies data entry	Consumes screen space
Requires modest training	
Gives convenient assistance	
Permits use of form-management tools	
Command language	
Flexible	Poor error handling
Appeals to "power" users	Requires substantial training and memorization
Supports user initiative	
Allows convenient creation of user-defined macros	
Natural language	
Relieves burden of learning syntax	Requires clarification dialog
	May not show context
	May require more keystrokes
	Unpredictable

The 8 golden rules of interface design

1. Strive for consistency
2. Cater to universal usability
3. Offer informative feedback
4. Design dialogs to yield closure
5. Prevent errors
6. Permit easy reversal of actions
7. Keep users in control
8. Reduce short term memory load



ACTIVITY 3: UI Audit: The Eight Golden Rules in the Wild

- Instructions:
  - Provide a brief, original summary (1–2 sentences per rule) explaining each of the Eight Golden Rules in your own words
- Select one website, desktop software, mobile app, or physical device display that you personally use. Identify a scenario where the interface violates at least two (2) of the Golden Rules. For your chosen example, document the following:
  - Interface Name & Context: What application/device is it, and what task were you trying to complete?
  - Visual Evidence: Include a screenshot, photo, or detailed walkthrough of the problem area.
  - Rule Violations: Identify the specific rules broken and explain how the design fails the user.
  - Redesign Proposal: Briefly explain how you would fix the design to satisfy the rules.

Prevent errors

- Make error messages specific, positive in tone, and constructive
- Mistakes and slips (Norman, 1983)
- Correct actions
  - Gray out inappropriate actions
  - Selection rather than freestyle typing
  - Automatic completion
- Complete sequences
  - Single abstract commands
  - Macros and subroutines



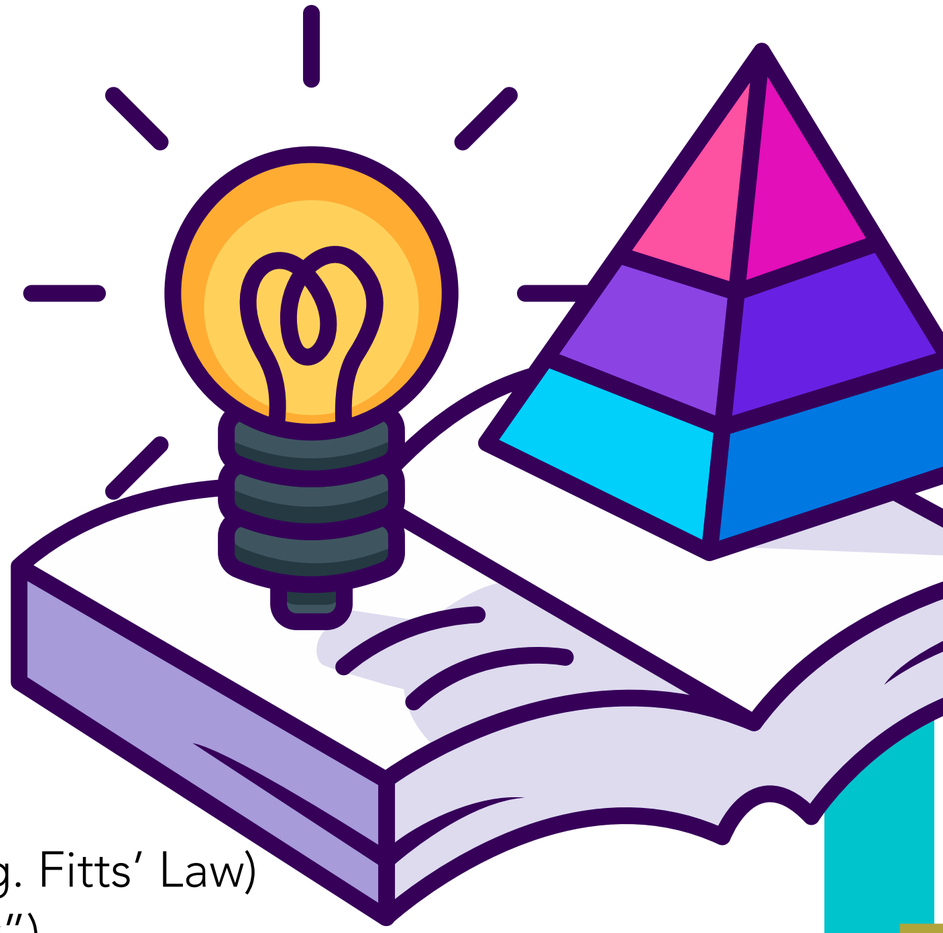
Theories

- Theories in design serve as comprehensive explanations or models that help designers understand user behavior, cognitive processes, and the interaction between users and systems. They offer deeper insights into why certain designs work or fail.



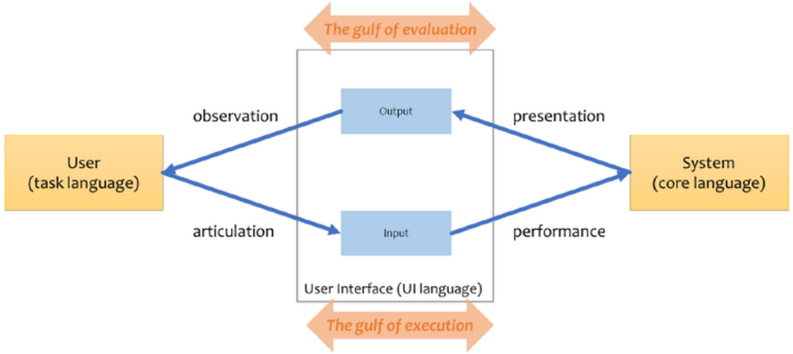
Theories

- Beyond the specifics of guidelines
- Types of theories
  - descriptive (describe user interfaces & their uses)
  - Explanatory - describe sequences of actions
  - Prescriptive – offer guidelines to make decisions
  - Predictive – offer predictions, e.g. on speed or errors
- Some theories are based on human capacity
  - Motor – pointing, clicking, dragging, other movements (e.g. Fitts' Law)
  - Perceptual – visual, auditory, tactile, other ("Gestalt Theory")
  - Cognitive – problem solving with short and long-term memory



Stages of action theory (Explanatory)

- **Norman's seven stages of action model**
  1. Forming the goal
  2. Forming the intention
  3. Specifying the action
  4. Executing the action
  5. Perceiving the system state
  6. Interpreting the system state
  7. Evaluating the outcome
- **Norman's contributions**
  - Context of cycles of action and evaluation.
  - ***Gulf of execution:*** Mismatch between the user's intentions and the allowable actions
  - ***Gulf of evaluation:*** Mismatch between the system's representation and the users' expectations



Consistency theories (Prescriptive)

Consistent user interface through grammar

- Definition is elusive – multiple forms sometimes in conflict
- Inconsistencies might be used (sparingly!) for drawing attention

Inconsistent action verbs

- Take longer to learn, cause more errors, slow down users, and are harder for users to remember

Consistent	**Inconsistent A**	**Inconsistent B**
delete/insert character	delete/insert character	delete/insert character
delete/insert word	remove/bring word	remove/insert word
delete/insert line	destroy/create line	delete/insert line
delete/insert paragraph	kill/birth paragraph	delete/insert paragraph

Consistency theory

- **Consistency** of nouns (objects) and verbs (actions)
  - Reduces learning time and errors
- Consistency of
  - Color
  - Layout
  - Icons
  - Fonts and Font sizes
  - Button sizes

Fitts' Law (in HCI)

- This theory predicts that the time required to move to a target area (e.g., clicking a button) depends on the distance to and size of the target. This helps designers place frequently used buttons closer to users and make them larger for easier interaction.

MID FIDELITY PROTOTYPE